1616 SMD Dual Flash LED Subsystem

Hardware Architecture, Bill of Materials (BOM) & Layout Specification

Document ID: SPEC-LED-1616-V1
Revision: 1.0 (Production)
Footprint: 1616 SMD (3.9×3.9 mm)

1. COMPLETE BILL OF MATERIALS (BOM)

Item	Designator	Manufacturer Part Number	Distributor PN (DigiKey / Mouser)	Qty	Description / Specifications
1	D1	`LUW F8DN-SGSI-RS-0-1000-R18-Z`	475-LUWF8DN-SGSI-RS-ND 720-LUWF8DN-SGSI-RS	1	ams OSRAM Cool White (6000K) 1616 SMD Flash LED, 1.0A Nom / 2.5A Peak
2	D2	`LUW F8BN-5B7B-1-1000-R18-Z`	475-LUWF8BN-5B7B-ND 720-LUWF8BN-5B7B	1	ams OSRAM Warm White (3000K) 1616 SMD Flash LED, 1.0A Nom / 2.5A Peak
3	U1	`LM3644TTYFFR`	36092540147365858885342458700869022 93	1	TI Dual 1.5A Flash LED Driver IC, I2C Interface, 12-DSBGA Package
4	L1	`TFM201610GHM-1R0MTAA`	445-175586-1-ND	1	TDK 1.0 µH Power Inductor, Shielded, Low-Profile (2016 Package)
5	C1, C2	`GRM188R60J226MEA0D`	490-13247-1-ND	2	Murata 22 µF, 6.3V, X5R MLCC Capacitor (0603 Package)
6	C3	`CL10A106KP8NNNC`	1276-1049-1-ND	1	Samsung 10 µF, 10V, X5R MLCC Capacitor (0603 Package)
7	R1, R2	`RC0402FR-0710KL`	311-10.0KLRCT-ND	2	Yageo 10 kΩ 1% Resistor (0402 Package) - I2C Pull-Up
8	R3	`RC0402FR-07100KL`	311-100KLRCT-ND	1	Yageo 100 kΩ 1% Resistor (0402 Package) - Strobe Pull-Down

2. PCB LAYOUT & THERMAL THERMAL ARCHITECTURE (CADENCE / ALLEGRO)

- **Thermal Via Array:** Place a 3×3 matrix of 0.25 mm thermal vias inside each central 1616 thermal pad. Specify IPC-4761 Type VII (vias filled and capped).
- **Copper Weight:** Use minimum 2 oz (70 µm) outer copper. Do not use thermal relief spokes on LED ground pads; tie directly to solid ground fills.
- **Trace Sizing:** Power rails (VIN, VOUT, SW, LED1, LED2) carrying up to 2.5A must be routed with minimum 1.0 mm (40 mils) trace width.

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1616 FOOTPRINT & THERMAL VIA MATRIX +----------
-------------------------+ | 1616 SMD Package
| | [Anode (+)] [Thermal] [Cathode] | +------+-
----------+-----------+-----+ | (3x3 Via Array)
| [Top Cu Pour] =====[GND]==== [Top Cu Pour] |
| | | | =====+=====+=====+=====+=====+===== 2
oz Bottom Cu
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3. ELECTRICAL CHARACTERISTICS & CIRCUIT INTEGRATION

Parameter	Operating Condition / Test State	Min	Typ	Max	Unit
Input Operating Voltage (V_{IN})	Connected to main system supply rail	2.7	3.6	5.5	V
Flash Pulse Current (I_{F_FLASH})	Hardware Strobe pulse duration ≤ 100 ms	1.0	1.5	2.5	A
Torch Mode Current (I_{F_TORCH})	Continuous video illumination mode	50	100	250	mA
Forward Voltage (V_F)	Per LED diode @ I_F = 1.0A	3.0	3.4	3.8	V
Color Temperature Tuning	Dual-channel ratio mixing (D1 Cool + D2 Warm)	2700	—	6500	K

4. SECONDARY OPTICS COMPATIBILITY

- **TIR Lens Array (40°–70°):** Recommended for standard field-of-view camera matching with high light efficiency.
- **Micro-Fresnel Diffuser:** Provides directional focus control with high center-spot intensity (15°–30° beam width).
- **Silicone Dome Diffuser:** Delivers ultra-soft 180° portrait lighting, eliminating harsh skin hot-spots.